

PIYUSH KUMAR

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CAREER OBJECTIVE

AI undergraduate with strong foundations in **Python**, **Data Structures**, and algorithmic problem solving. Experienced in building backend systems, AI-powered applications, and vector database solutions, with hands-on expertise in **FastAPI**, **PostgreSQL**, **JWT Authentication**, and Retrieval-Augmented Generation (RAG).

PROJECTS

VectorForge — AI Vector Database & RAG Engine

May 2026 – June 2026

- Developed a vector database and semantic search engine for high-dimensional embedding storage, document retrieval, and AI-powered question answering.
- Built in **Python** using **FastAPI**, implementing **HNSW**, **KD-Tree**, and **Brute Force** search algorithms with integration of **Ollama**, **nomic-embed-text**, and **llama3.2** for Retrieval-Augmented Generation (RAG).
- Designed to demonstrate the internal workings of modern AI retrieval systems by enabling efficient semantic search, vector indexing, and context-aware response generation through local LLMs.

Nyay – Legal Case Management App

March 2026 – April 2026

- Developed a multi-role Android application for digital legal case filing and case lifecycle management.
- Built using **Kotlin**, **Jetpack Compose**, and **Firebase Firestore**, implementing real-time workflow tracking and a structured 5-step case handling system with document upload support.
- Created to simplify and digitize legal case management by improving workflow transparency, document organization, and status tracking for different user roles.

HabitFlow — Habit Tracking REST API

July 2026 – August 2026

- Developed a secure RESTful backend for habit tracking, enabling user authentication, habit management, progress tracking, and dashboard analytics.
- Built using **FastAPI**, **PostgreSQL**, **SQLAlchemy**, and **JWT Authentication**, implementing protected CRUD APIs, password hashing with **bcrypt**, and database-driven user workflows.
- Designed to demonstrate scalable backend development practices through secure authentication, relational database modeling, RESTful API architecture, and habit analytics.

SKILLS

Languages: Python, SQL, Kotlin **Backend:** FastAPI, PostgreSQL, SQLAlchemy, JWT Authentication, REST APIs **AI:** RAG, Vector Databases, Semantic Search, NLP **Tools:** Git, Ollama, Jetpack Compose

CERTIFICATIONS

- **HackerRank** — Python Programming [\[Certificate\]](#)
- **Infosys Springboard** — Programming Fundamentals using Python [\[Certificate\]](#)

KEY ACHIEVEMENTS

- Built backend systems using **FastAPI**, **PostgreSQL**, and **JWT Authentication**.
- Solved **150+ coding problems** across **LeetCode**, **GeeksforGeeks**, and **HackerRank**.
- Developed AI-powered applications involving **Vector Databases**, **Semantic Search**, and **Retrieval-Augmented Generation (RAG)**.

EDUCATION

B.Tech in Artificial Intelligence

Expected 2028

AKTU, Lucknow